ChemCorp Industries S.A.
ANALYTICAL DATASETS

Energy and Environmental Consumption Data 2024
Ref: DS-ENERGIE-2024

DESCRIPTION

This dataset presents the monthly consumption of energy (electricity, natural gas, steam), water and waste generation per production unit at the Villeurbanne site for the year 2024. These data feed the annual ESG report and the carbon reporting (scopes 1 and 2).

ANALYTICAL METHODS AND ASSUMPTIONS

Electricity: Schneider PowerLogic PM8000 sub-meters per unit — daily readings, aggregated monthly. Emission factor: 0.0567 kgCO■/kWh (ADEME 2024 — French grid).

Natural gas: Elster BK-G16 meters — monthly readings. Emission factor: 0.185 kgCO■/kWh NCV (ADEME 2024).

Steam: Yokogawa ADMAG mass meters on the distribution line. Steam emission factor: 0.070 kgCO■/kg (in-house gas boiler production).

Water: Endress+Hauser Prosonic Flow ultrasonic meters on the demineralised water and process water networks.

Waste: Waste tracking forms (BSD) CERFA No. 12571. Hazardous waste accounted for separately.

CO■ equivalent: Calculated using the formula: CO■ = Elec×0.0567 + Gas×0.185 + Steam×0.070 (in tonnes). Scope: scopes 1 and 2 only.

SPECIFICATIONS / REFERENCE VALUES

Indicator	2022	2023	2024 target	2027 target
Total electricity (MWh)	8,420	8,102	7,454 (-8%)	6,320 (-22%)
Total natural gas (MWh NCV)	2,845	2,710	2,493 (-8%)	2,140 (-21%)
Total process water (m³)	48,200	45,600	43,300 (-5%)	36,500 (-20%)
Total waste (t)	812	745	690 (-7%)	520 (-30%)
Total CO■ (t eq.)	1,247	1,198	1,102 (-8%)	868 (-28%)
CO■ intensity (kgCO■/t product)	1.39	1.31	1.20 (-8%)	0.97 (-26%)

DS-ENERGIE-2024 — Raw data

Month	Unit	Elec. (MWh)	Gas (MWh NCV)	Steam (t)	Water (m³)	Waste (t)	CO_2 (t eq.)
Jan	Solvents workshop	163.2	36.0	13.5	42.8	2.36	16.86
Jan	Acids workshop	108.2	61.8	30.9	60.7	3.75	19.73
Jan	Bases workshop	75.4	87.8	50.7	47.3	2.03	24.07
Jan	Specialities workshop	113.9	18.3	8.1	29.5	4.27	10.41
Jan	Utilities / HVAC	98.8	49.4	72.9	134.7	0.56	19.84
Jan	Lighting / Misc.	23.6	5.4	2.1	16.1	0.32	2.48
Feb	Solvents workshop	158.4	35.0	13.1	41.5	2.29	16.37
Feb	Acids workshop	105.7	60.4	30.2	59.3	3.67	19.28
Feb	Bases workshop	67.9	79.0	45.6	42.6	1.82	21.66
Feb	Specialities workshop	117.3	18.9	8.4	30.4	4.4	10.74
Feb	Utilities / HVAC	97.3	48.6	71.9	132.7	0.55	19.54
Feb	Lighting / Misc.	23.0	5.2	2.1	15.7	0.31	2.41
Mar	Solvents workshop	146.5	32.3	12.1	38.4	2.12	15.13
Mar	Acids workshop	105.9	60.5	30.3	59.4	3.67	19.32
Mar	Bases workshop	64.0	74.5	43.0	40.1	1.72	20.42
Mar	Specialities workshop	117.4	18.9	8.4	30.4	4.4	10.74
Mar	Utilities / HVAC	87.6	43.8	64.7	119.4	0.5	17.6
Mar	Lighting / Misc.	22.9	5.2	2.1	15.6	0.31	2.41
Apr	Solvents workshop	159.2	35.1	13.2	41.7	2.31	16.44
Apr	Acids workshop	100.6	57.5	28.8	56.5	3.49	18.36
Apr	Bases workshop	68.9	80.2	46.3	43.2	1.85	21.98
Apr	Specialities workshop	119.6	19.2	8.5	31.0	4.49	10.93
Apr	Utilities / HVAC	90.8	45.4	67.0	123.8	0.52	18.24
Apr	Lighting / Misc.	22.3	5.1	2.0	15.2	0.3	2.35
May	Solvents workshop	158.6	35.0	13.1	41.6	2.3	16.38
May	Acids workshop	101.3	57.9	29.0	56.9	3.52	18.49
May	Bases workshop	69.4	80.8	46.6	43.5	1.87	22.14
May	Specialities workshop	121.1	19.5	8.7	31.4	4.54	11.08
May	Utilities / HVAC	89.9	44.9	66.4	122.5	0.51	18.05
May	Lighting / Misc.	22.7	5.2	2.1	15.5	0.31	2.4
Jun	Solvents workshop	140.6	31.0	11.6	36.8	2.04	14.52
Jun	Acids workshop	103.2	59.0	29.5	57.9	3.58	18.83
Jun	Bases workshop	65.6	76.4	44.1	41.1	1.76	20.94
Jun	Specialities workshop	112.6	18.1	8.0	29.2	4.22	10.29
Jun	Utilities / HVAC	85.8	42.9	63.4	117.0	0.49	17.24
Jun	Lighting / Misc.	21.9	5.0	2.0	14.9	0.3	2.31
Jul	Solvents workshop	146.9	32.4	12.2	38.5	2.13	15.18
Jul	Acids workshop	94.8	54.2	27.1	53.2	3.29	17.3
Jul	Bases workshop	61.8	71.9	41.5	38.7	1.66	19.71
Jul	Specialities workshop	118.0	19.0	8.4	30.5	4.42	10.79
Jul	Utilities / HVAC	82.4	41.2	60.8	112.3	0.47	16.55
Jul	Lighting / Misc.	22.1	5.0	2.0	15.1	0.3	2.32

Month	Unit	Elec. (MWh)	Gas (MWh NCV)	Steam (t)	Water (m^3)	Waste (t)	CO■ (t eq.)
Aug	Solvents workshop	142.9	31.5	11.8	37.5	2.07	14.76
Aug	Acids workshop	101.3	57.9	28.9	56.9	3.52	18.48
Aug	Bases workshop	64.0	74.5	43.0	40.1	1.72	20.42
Aug	Specialities workshop	107.9	17.3	7.7	27.9	4.05	9.86
Aug	Utilities / HVAC	90.2	45.1	66.6	123.0	0.51	18.12
Aug	Lighting / Misc.	22.7	5.2	2.1	15.5	0.31	2.4
Sep	Solvents workshop	147.4	32.5	12.2	38.6	2.13	15.22
Sep	Acids workshop	103.5	59.2	29.6	58.1	3.59	18.89
Sep	Bases workshop	71.0	82.7	47.7	44.5	1.91	22.66
Sep	Specialities workshop	109.4	17.6	7.8	28.3	4.1	10.0
Sep	Utilities / HVAC	89.5	44.7	66.1	122.0	0.51	17.97
Sep	Lighting / Misc.	22.0	5.0	2.0	15.0	0.3	2.31
Oct	Solvents workshop	159.7	35.2	13.2	41.8	2.31	16.49
Oct	Acids workshop	95.2	54.4	27.2	53.4	3.3	17.37
Oct	Bases workshop	71.4	83.1	48.0	44.8	1.92	22.78
Oct	Specialities workshop	120.2	19.3	8.6	31.1	4.51	10.99
Oct	Utilities / HVAC	89.2	44.6	65.9	121.6	0.51	17.92
Oct	Lighting / Misc.	23.7	5.4	2.2	16.1	0.32	2.5
Nov	Solvents workshop	149.3	32.9	12.4	39.1	2.16	15.42
Nov	Acids workshop	107.4	61.4	30.7	60.3	3.73	19.6
Nov	Bases workshop	75.1	87.4	50.4	47.1	2.02	23.96
Nov	Specialities workshop	110.9	17.8	7.9	28.7	4.16	10.13
Nov	Utilities / HVAC	94.5	47.3	69.8	128.9	0.54	18.99
Nov	Lighting / Misc.	23.6	5.4	2.1	16.1	0.32	2.48
Dec	Solvents workshop	150.7	33.3	12.5	39.5	2.18	15.58
Dec	Acids workshop	105.3	60.2	30.1	59.1	3.65	19.21
Dec	Bases workshop	68.4	79.7	46.0	42.9	1.84	21.84
Dec	Specialities workshop	119.8	19.3	8.6	31.0	4.49	10.97
Dec	Utilities / HVAC	97.6	48.8	72.1	133.2	0.55	19.61
Dec	Lighting / Misc.	23.3	5.3	2.1	15.9	0.32	2.45

DS-ENERGIE-2024 — Statistical summary

DESCRIPTIVE STATISTICS

Subset	Parameter	Observed range	Mean	Std dev.	Specification
Solvents workshop	Elec. (MWh/month)	132–165	148	10	OEE >80%
Acids workshop	Gas (MWh/month)	48–68	57	7	Boiler efficiency
Site total	CO■ per month (t)	82–128	99	13	Annual target 1,102 t
Bases workshop	Water (m^3/month)	35–55	43	6	Target -5%/year
Site total	Waste (t/month)	55–72	62	5	Target -7% in 2024

OBSERVATIONS AND CONCLUSIONS

• Taken together, the results confirm that the production process remained in statistical control over the period considered.

• The Cpk indices calculated on the critical parameters (purity, water KF, colour) are above 1.33 for all 3 products, indicating satisfactory process capability.

• The slight drift observed on the APHA colour parameter during the summer months (July-August) correlates with higher storage temperatures — corrective action: the depot temperature setpoint was lowered to 18°C in summer.

• No definitively non-conforming batch was released to a customer. The 3 batches placed under review were successfully reprocessed.

• The data are archived in the ChemCorp LIMS (QC-TRACK v4.2 module) and available for COFRAC audit.

Data generated by ChemCorp LIMS v4.2 — DS-ENERGIE-2024 — Confidentiality: internal. Contact: qualite@chemcorp.fr